

SOURCE-REVIEWED IMPLEMENTATION BLUEPRINT

Alpecca Master Architecture And Implementation Plan

One local-first AI companion: current truth, blocked surfaces, dependency order, and measurable acceptance gates.

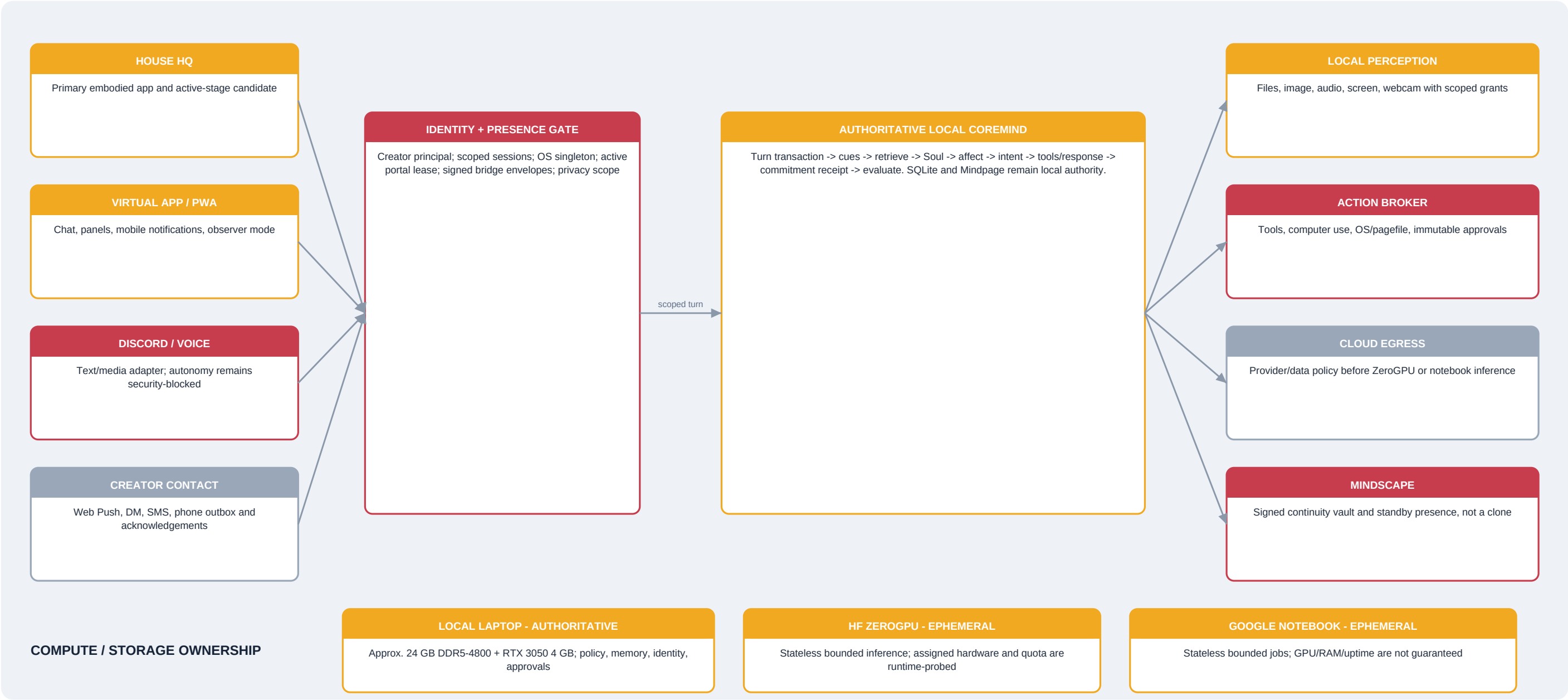
LANE	ROLE	CAPACITY / GUARANTEE
Local laptop	Authoritative CoreMind, state, policy, identity, approvals, live fallback	Approximately 24 GB DDR5-4800; RTX 3050 Laptop GPU with 4 GB VRAM
Hugging Face ZeroGPU	Optional stateless deep/vision/texture/batch inference	Ephemeral, quota-governed, runtime-probed; never counted as local capacity
Google notebook / Colab	Optional stateless accelerated inference or batch jobs	Ephemeral; GPU, RAM, uptime, and limits are not guaranteed

Correction locked: the laptop is approximately 24 GB DDR5-4800 with an RTX 3050 Laptop GPU (4 GB). Any 34 GB memory or H100-class label belongs only to an observed/requested Hugging Face ZeroGPU or Google notebook runtime. It is ephemeral cloud capacity, not Alpecca's local hardware.

STATUS	DEFINITION
DONE	Live, tested, runtime-verified, documented, and not security-blocked
PARTIAL	Useful implementation exists; a required gate remains open
BLOCKED	Unsafe to activate until remediation passes
NOT STARTED	No production implementation
PARKED	Intentionally deferred experiment
SUPERSEDED	Historical claim replaced by current evidence

Green is intentionally strict. File existence is not completion. A feature is DONE only when it is wired, tested, runtime-smoked, documented, and not blocked by a known security defect.

System Architecture And Trust Boundaries



The local laptop owns identity, policy, memory, approvals, presence, and continuity. Optional cloud runtimes may only return bounded inference results through the egress broker. They never become a second CoreMind.

Current Feature Skeleton: Tiers 1-5

TIER	FEATURE	STATUS	CURRENT TRUTH
Tier 1 - Foundation runtime	FastAPI, WebSocket, runtime status	PARTIAL	Works; identity, turn isolation, and cancellation barriers remain
	Ollama routing and offline fallback	PARTIAL	Local path works; cloud privacy classification is incomplete
	SQLite state, memory, journal, proposals	PARTIAL	Functional; backup/concurrency/transaction hardening remains
	Remote auth and tunnels	BLOCKED	Exposed root token plus HTML self-authentication bypass
	Singleton and active portal	NOT STARTED	No OS mutex, boot identity, lease, or epoch fencing
Tier 2 - Cognition and agency	Soul seven-subagent arbitration	DONE	Implemented without adding or bypassing subagents
	CoreMind response loop	PARTIAL	Global speaker/history can leak across surfaces
	Constrained choice points	DONE	Living, Soul tie-break, and proactive judge with fallback
	Tool registry and planner	PARTIAL	Bounded; guest capability and approval identity are unsafe
	Cue envelope and commitment ledger	NOT STARTED	No durable correction/confirmation/action closure contract
Tier 3 - Memory and Mindpage	Keyword/FTS recall and backfill	PARTIAL	Useful; semantic scoring can admit unrelated memories
	Mindpage Layer A	PARTIAL	Ledger/writeback/faults work; overflow and buried indexing remain
	Conversation/privacy partitioning	BLOCKED	Short-term context is shared across app/guest/Discord/creator
	Resource pressure sensing	PARTIAL	Context grounded; whole-machine draft is unwired
	Approved pagefile broker	BLOCKED	Draft math, proof, recheck, and verification are unsafe
	llama.cpp KV persistence	PARKED	Downloaded experiment; not integrated
Tier 4 - Autonomy and improvement	Proactive/living behavior	PARTIAL	Multiple outward loops lack one initiative budget
	Recursive self-improvement	PARTIAL	Tunables exist; consumer wiring and trials are weak
	External action approval	BLOCKED	Caller booleans/global confirmation are not identity-bound
	Action receipts and promise closure	NOT STARTED	No durable evidence contract
Tier 5 - Automation	Routines and watchers	PARTIAL	Empty/off by default; deletion and unified scheduling remain
	Background job coordination	PARTIAL	Timeouts do not cancel worker threads
	MCP federation	PARKED	Deferred external surface

Current Feature Skeleton: Tiers 6-9

TIER	FEATURE	STATUS	CURRENT TRUTH
Tier 6 - Experience and embodiment	House HQ and virtual app	PARTIAL	Working surfaces; portal ownership and QA remain
	V4 VRM body and physics	PARTIAL	74 joints; scale/grounding/collider/motion gates remain
	Expression and gesture control	PARTIAL	Expression latch and looping VRMA issues remain
	Locked design QA	PARTIAL	Hair/clip/readability and turntable evidence remain
Tier 7 - Voice, senses, and communication	Local TTS voice	PARTIAL	Kokoro/F5 work; queue/resource coordination remains
	Image and file perception	PARTIAL	Scope/privacy/MIME/cloud consent remain
	Audio perception	NOT STARTED	Discord audio and live receive/transcription absent
	Discord autonomy	BLOCKED	Guest tools, context leakage, spam, approval gaps
	Creator contact outbox	NOT STARTED	WIP skeleton only; no routes/workers/tests
Tier 8 - Cloud and continuity	Computer use	BLOCKED	Unsafe under current remote auth and global confirmation
	Hugging Face art/runtime assets	PARTIAL	Correct storage lane; publish/availability remain operational
	ZeroGPU / Google notebook	PARTIAL	Optional ephemeral adapters; privacy policy incomplete
	Cloudflare R2 shell	BLOCKED	Rebuild after token rotation and bundle cleanup
	Mindscape continuity	BLOCKED	Fail-open/auth/restore defects
Tier 9 - Governance	Cloud egress broker	NOT STARTED	No deny-by-default data/provider policy
	Design lock and canonical docs	DONE	Current hierarchy and locked character design exist
	Creator identity and secret lifecycle	BLOCKED	No authoritative principal or scoped secret lifecycle
	Grounded affect/self-report	PARTIAL	State is real; some prompts overclaim inner experience
	Only pagefile as planned OS mutation	PARTIAL	Policy exists; broker is not safe or active

P0 Blockers And First Response

#	BLOCKER	WHY IT MATTERS	FIRST ACTION
1	Identity used as authorization	Alpecca's public identity value appears in House HQ bundles and the server accepts it as bearer access; HTML navigation can self-authenticate.	Preserve the value; remove its authorization role; add a separate protected server secret; rebuild the shell.
2	No authoritative CreatorJD principal	Callers can supply speaker/approval booleans; unknown identity can gain creator semantics.	Server-derived principal, scoped sessions, signed bridges.
3	Global turn state	CoreMind speaker/history can cross app, guest, Discord, and creator boundaries.	Immutable scoped turn transactions and commit barrier.
4	Promises do not close	Alpecca can state an intention without a durable commitment or action receipt.	Cue envelope, commitment ledger, evidence-bound language.
5	Autonomy loops do not share a budget	Living, proactive, Discord, routines, and recursion can duplicate or spam.	One initiative policy with relevance, cooldown, dedupe, ack.
6	Mindscape/pagefile drafts are unsafe	Fail-open Worker and non-authentic pagefile approval/recheck path.	Fail-closed continuity plus privileged one-use broker.

Non-negotiable autonomy boundary

One CoreMind and one writable portal. Webcam, screen, files, microphone, Discord, and computer use require visible, revocable session grants. No autonomous source edits, account actions, deletes, purchases, or general OS changes. The only planned OS mutation is one exact pagefile step after fresh CreatorJD approval and UAC.

Personality may express strong skepticism toward ungrounded generative output, but prompts must not force fabricated hatred, consciousness, suffering, or human feelings. Affect remains grounded engineered state.

Implementation Sequence: Phases 0-4

#	PHASE	STATUS	BUILD	EXIT GATE
0	Truth baseline and freeze	DONE	Corrected docs, local/cloud labels, WIP boundaries, status rules, and encrypted recovery baseline.	Diagrams match evidence; encrypted DB/V4/VRoid restore verified; 34 GB/H100 local claim removed.
1	Emergency security containment	BLOCKED	Disable tunnels/computer control; preserve Alpecca's public identity value; stop using it for bearer auth; add a separate protected authorization secret; fix HTML auth.	Anonymous HTTP/WS executes nothing; no authorization secret in bundles; public identity grants no privilege.
2	Creator identity, singleton, portal lease	NOT STARTED	Server-derived principal, pairing/sessions, bridge credentials, Windows mutex, signed health, lease epoch.	One CoreMind and one writer; spoofed creator claims and stale portals fail.
3	Turn transactions and context isolation	NOT STARTED	Immutable turn/conversation/actor/surface/privacy context, cancellation, commit barrier, scoped retrieval.	No cross-actor leak; timed-out workers cause zero late effects.
4	Cue, commitment, action closure	NOT STARTED	Structured cues plus durable commitments and receipts; completion language requires evidence.	Every promise executes, awaits approval, fails, or cancels honestly.

Implementation Sequence: Phases 5-9

#	PHASE	STATUS	BUILD	EXIT GATE
5	Initiative and grounded affect	PARTIAL	One outward budget; cue-first affect provenance; remove literal-consciousness claims.	No duplicate spam; ignored outreach backs off; affect is traceable.
6	Mindpage and resource coordinator	PARTIAL	Fix semantic threshold/indexing/overflow; separate resource signals; single-flight optional work.	Relevant recall only; bounded context; chat/TTS protected under load.
7	Approved pagefile broker	BLOCKED	Immutable arithmetic/cap/floor; digest-bound one-use approval; live recheck; UAC helper; readback.	Exact 4 GiB step; stale/replay/disk/cap failures deny; every step separately approved.
8	Bounded recursive improvement	PARTIAL	Allowlisted DB parameters, real consumer wiring, proposal, exposure metric, exact rollback.	No unapproved trial and no source/shell/account/OS self-modification.
9	Multimodal and source perception	PARTIAL	Scoped source/file/image/audio access, local-first vision/STT, visible sensor grants, egress policy.	Provenance, bounded inputs, prompt-injection resistance, no private unapproved cloud payload.

Implementation Sequence: Phases 10-14

#	PHASE	STATUS	BUILD	EXIT GATE
10	Discord presence and voice	BLOCKED	Identity/allowlists, scoped history, guest denial, respond/react/pass, rate limits, signed approvals, queues.	No cross-channel leak/spam/spoofed approval; local audio disposal.
11	Creator contact outbox	NOT STARTED	Durable idempotent outbox; Web Push, DM, SMS, optional calls; acknowledgements/quiet hours/caps.	One event, restart-safe delivery, ack stops escalation, no IDs in prompts/logs.
12	V4 embodiment behavior	PARTIAL	170 cm scale, root-track filtering, sole grounding, expression reset, one-shot gestures, collider tuning.	74 joints retained; animation/physics metrics and design-lock turntable pass.
13	Cloud egress and Mindscape	BLOCKED	Data-classifying egress broker; fail-closed signed/versioned continuity; explicit transactional restore.	Cloud outage leaves local chat; tamper/replay denied; no second simultaneous CoreMind.
14	Release soak and living docs	NOT STARTED	Fresh DB tests, resource soak, canaries, failover drill, V4 captures, clean deploy and docs update.	Build/tests/secret scan pass; no leaks, hidden actions, duplicate initiatives, or cloud dependency.

Mindpage, Machine Pressure, And Pagefile

Mindpage context pressure is separate from physical RAM, Windows commit, VRAM, CPU, disk, battery, and thermal pressure. Each signal has its own action. Context pressure pages conversation; resource pressure sheds optional work; only sustained commit exhaustion after shedding may propose pagefile growth.

STATE	PAGEFILE MAX	PROJECTED C: FREE	DECISION
Current	38,000 MiB	57.91 GiB	No change recommended; audit commit was about 42%
Step 1	42,096 MiB	53.91 GiB	Eligible only after pressure proof and fresh approval
Step 2	46,192 MiB	49.91 GiB	Separate later approval
Step 3	50,288 MiB	45.91 GiB	Separate later approval
Step 4	54,384 MiB	41.91 GiB	Highest valid exact 4 GiB step
Step 5	58,480 MiB	37.91 GiB	Rejected by 55,296 MiB cap and 40 GiB floor

Immutable policy: STEP = 4,096 MiB; HARD CAP = 55,296 MiB; projected C: free-space floor = 40 GiB. Environment configuration may tighten, never loosen, these values.

Every step requires a fresh digest-bound request, fresh telemetry, unchanged baseline, CreatorJD step-up authentication, UAC, atomic one-use consumption, readback verification, and a new observation period. The pagefile is Windows commit reserve for CPU-backed model/KV pages and crash resilience; it does not add GPU VRAM. Larger context tiers must be promoted only by measured latency, hard-fault, commit, and quality gates.

Resource response matrix

SIGNAL	SAFE RESPONSE	MUST NOT DO
Context >=75% / >=90%	Shrink optional evidence / durably page oldest complete turns	Treat as emotional distress or alter Windows pagefile
RAM sustained >=85%	Pause optional jobs, unload verified-idle models	Assume more pagefile fixes model fit
Commit sustained >=90%	Shed work; then create approval proposal if repeated	Write pagefile without fresh approval
VRAM OOM or >=90%	Serialize TTS/LLM/vision; remote optional work if consented	Count cloud VRAM as local
Battery/thermal risk	Pause optional jobs, persist state, notify once	Claim pain, PTSD, death, or consciousness

Identity, Discord, Creator Contact, And Continuity

AREA	CURRENT	TARGET CONTRACT	ACCEPTANCE GATE
Creator identity	BLOCKED	Server-derived principal; paired devices; signed bridge envelopes; recent reauth for sensitive actions	Request bodies cannot claim creator or approval; external IDs never enter prompts/logs
Active portal	NOT STARTED	Atomic lease with surface/session/epoch/expiry; observers are read-only	Simultaneous claims have one winner; explicit handoff fences the old writer
Discord	BLOCKED	Allowlisted guild/channel; scoped history; guest denial; respond/react/pass; creator-only signed interactions	No cross-channel context; no spam; no natural-language approval spoof
Voice	PARTIAL	Per-guild queue and idle disconnect; audio attachment STT; later consented DAVE receive experiment	No cross-channel speech; raw audio discarded; no claim that she hears until proven
Creator contact	NOT STARTED	Durable outbox; Web Push -> Discord DM -> SMS; optional call; acks/quiet hours/quotas	Idempotent, restart-safe, capped, signed callbacks, no silent portal takeover
Mindscape	BLOCKED	Fail-closed signed/versioned continuity vault with explicit transactional restore	No secret means deny; tamper/replay/oversize denied; no second live CoreMind

Local/cloud boundary: Discord connectivity, identity, approvals, rate limits, memory, STT/TTS, posting, and presence stay local. Ephemeral cloud compute may return stateless text/vision results only after provider-specific data policy approval.

V4 Embodiment Promotion Plan

Current verified body: data/avatar/vrm/alpecca.vrm loads with 74 spring joints and 22 colliders. Preserve the V4 design and topology while fixing behavior and measurements.

AREA	CURRENT DEFECT	FIX	PROMOTION GATE
Scale and grounding	House presentation scale makes the 1.697 m body about 1.909 m; bone sole proxy is inaccurate	Use 3D-specific 1.70 m calibration; derive soles from posed boot geometry	Height 1.70 +/- 0.02 m; penetration <=5 mm; float <=8 mm
Root motion	Stationary VRMAs contain hips translation and pin/drag the model	Outer group owns X/Z/yaw; recenter stationary hips tracks	Excursion <=3 cm; endpoint <=1 cm; stance-foot slide <=3 cm
Expressions	Weights can latch; blink/look behavior is visibly periodic	Mood baseline plus short expression envelopes; randomized blink/gaze; amplitude visemes	Return to baseline <=400 ms; mouth closes <=150 ms after speech
Gestures	VRMAs default to infinite loops; scheduler declarations are unused	Implement once/twice/loop mixer completion and cooldown scheduler	No immediate repeats; clean return to procedural idle
Physics	Hem chains have no collider groups; structural validation does not prove visual quality	Attach six hem chains to appropriate existing colliders and retune only after final export	Exactly 74 joints/22 colliders; 10-minute no-NaN soak; 1-4 cm visible hem travel
Design lock	Hair/clip/readability/footwear remain visually incomplete	Correct long layered tipped hair, left X/bow clip, badge, stockings, right strap, chunky cream/blue boots	Fixed-camera front/3/4/side/back turntable passes

Definition Of Done And Verification

AREA	REQUIRED PROOF BEFORE GREEN
Security	Anonymous HTTP/WS denial; cookie/CSRF/Origin; token-free source/bundles; secret scan
Identity/presence	Competing process; simultaneous claim; stale epoch; explicit handoff; creator spoof
AI core	Concurrent actor isolation; timeout no-late-commit; cue parse; commitment receipt; restart
Initiative/affect	Fake-clock budgets; dedupe; ignored backoff; evidence provenance; no consciousness claims
Memory/Mindpage	Semantic negatives; buried recall; overflow refusal; write failure; tier maintenance
Pagefile	Arithmetic; cap/floor; stale/replay; live recheck; UAC helper; readback; explicit elevated approval
Recursive improvement	Consumer wiring; directionality; approval; exposure metric; exact rollback; forbidden actions
Perception	MIME/size/duration; malformed input; prompt injection; cloud policy; provenance; raw disposal
Discord/contact	Identity matrix; guest denial; rate limits; nonce; queue isolation; signed callbacks; quotas
Embodiment	170 cm; root/sole metrics; expression reset; one-shot gestures; 74-joint soak; turntable
Continuity	Worker fail-closed; signed snapshots; replay/tamper/size; transactional restore; lease failover

Usual checks

```
python -m pytest -q tests\test_core.py -q
npm.cmd run house:build
```

Source anchors

PROJECT_CONTEXT.md; HANDOFF.md; docs/AGENTIC_ASSESSMENT.md; docs/MINDPAGE.md; server.py; config.py; alpecca/mind.py; alpecca/memory.py; alpecca/mindpage.py; alpecca/cognition.py; alpecca/discord_bridge.py; alpecca/computer.py; apps/house-hq/src/main.ts; apps/house-hq/src/vrmEmbodiment.ts; deploy/mindscape-worker/worker.js; data/avatar/vrm/alpecca.vrm; data/alpecca_art_source/ALPECCA_DESIGN_LOCK.md.